

# Reading a Line Plot with Fractions

## Answer Key

Grades 4–5 Mathematics

### Quick Answers

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|--------------------|-----------------------------------------|
| 1. 4               | 7. $1 \cdot \frac{1}{2}$ L              |
| 2. 16              | 8. $\frac{3}{8}$ L                      |
| 3. $\frac{3}{8}$   | 9. $\frac{27}{4}$ L                     |
| 4. $\frac{1}{4}$ L | 10. $\frac{3}{4}$ L                     |
| 5. $\frac{1}{2}$ L | 11. 4                                   |
| 6. $\frac{1}{2}$ L | 12. (a) new median = $\frac{3}{8}$ L, — |
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### What is verified

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11 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 12. The worked solution states what a correct response must contain.

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### Curriculum

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**Level:** Grades 4–5 Mathematics

5.MD – *problems 1, 2, 3, 5, 8, 11, 12*

5.NF.A.1 – *problems 4, 6, 7, 9, 10*

*Difficulty 1–5 of 5 across 13 tagged checks.*

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1. How many beakers hold exactly  $\frac{3}{8}$  of a litre?

**Solution:** Find the tick labelled  $\frac{3}{8}$  and count the crosses stacked above it. There are four.

4

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2. How many beakers are recorded on the plot altogether?

**Solution:** Add every stack, left to right:  $2 + 3 + 4 + 3 + 2 + 2$ .

16

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3. Which amount was recorded more often than any other?

**Solution:** Compare the heights of the stacks, not the amounts underneath them: the tallest stack has four crosses, one more than the next tallest.

$\frac{3}{8}$

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4. The two beakers at  $\frac{5}{8}$  of a litre are emptied into one jug. How much water is in the jug?

**Solution:** Two equal amounts added is the amount doubled:  $2 \times \frac{5}{8} = \frac{10}{8}$ , which reduces to  $\frac{5}{4}$ .

$1\frac{1}{4}$  L

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5. What is the difference between the greatest amount and the least amount on the plot?

**Solution:** The leftmost cross sits at  $\frac{1}{8}$  and the rightmost at  $\frac{3}{4}$ . Rewrite over eighths and subtract:  $\frac{6}{8} - \frac{1}{8}$ .

$\frac{5}{8}$  L

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6. How much water is held altogether by the beakers whose amounts are less than  $\frac{1}{2}$  of a litre?

**Solution:** Three stacks sit left of  $\frac{1}{2}$ : two beakers at  $\frac{1}{8}$ , three at  $\frac{1}{4}$ , four at  $\frac{3}{8}$ .

Work in eighths:  $2 \times \frac{1}{8} = \frac{2}{8}$ ,  $3 \times \frac{1}{4} = \frac{6}{8}$ ,  $4 \times \frac{3}{8} = \frac{12}{8}$ . Adding gives  $\frac{20}{8} = \frac{5}{2}$ .

$2\frac{1}{2}$  L

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7. The four beakers at  $\frac{3}{8}$  of a litre are poured together and then shared equally between 3 empty beakers. How much is in each?

**Solution:** Pour first:  $4 \times \frac{3}{8} = \frac{12}{8} = \frac{3}{2}$  litres in the jug.

Then share between 3:  $\frac{3}{2} \div 3 = \frac{3}{2} \times \frac{1}{3} = \frac{1}{2}$ .

$\frac{1}{2}$  L

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8. What is the median of the 16 recorded amounts?

**Solution:** The plot is already in order from left to right, so count crosses from the left. With 16 values the median sits between the 8th and the 9th.

Counting: crosses 1–2 are  $\frac{1}{8}$ , 3–5 are  $\frac{1}{4}$ , 6–9 are  $\frac{3}{8}$ . Both the 8th and the 9th cross sit at the same amount, so their midpoint is that amount.

$\frac{3}{8}$  L

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9. How much water is in the science cupboard altogether?

**Solution:** Multiply each amount by its stack height and add. In eighths throughout:

$$\begin{aligned} 2 \times \frac{1}{8} + 3 \times \frac{1}{4} + 4 \times \frac{3}{8} + 3 \times \frac{1}{2} + 2 \times \frac{5}{8} + 2 \times \frac{3}{4} \\ = \frac{2}{8} + \frac{6}{8} + \frac{12}{8} + \frac{12}{8} + \frac{10}{8} + \frac{12}{8} = \frac{54}{8} = \frac{27}{4} \end{aligned}$$

Written as a mixed number, that is

$6\frac{3}{4}$  L

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10. How much more water is held by all the beakers at  $\frac{1}{2}$  of a litre together than by all the beakers at  $\frac{1}{4}$  of a litre together?

**Solution:** Each group has three beakers. The half-litre group holds  $3 \times \frac{1}{2} = \frac{3}{2}$ ; the quarter-litre group holds  $3 \times \frac{1}{4} = \frac{3}{4}$ .

Subtract over quarters:  $\frac{6}{4} - \frac{3}{4}$ .

$\frac{3}{4}$  L

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**11.** How many beakers hold more than  $\frac{1}{2}$  of a litre?

**Solution:** “More than” excludes the stack sitting exactly at  $\frac{1}{2}$ . Count only the stacks to its right: two crosses at  $\frac{5}{8}$  and two at  $\frac{3}{4}$ , so  $2 + 2$ .

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**12.** One more beaker, holding  $\frac{7}{8}$  of a litre, is added. (a) What is the median of the 17 amounts now? (b) Explain why adding the largest amount did not move the median very far.

**Solution (a):** With 17 values the median is the 9th from the left. The new cross goes on the far right, so the first nine crosses are unchanged: 1–2 at  $\frac{1}{8}$ , 3–5 at  $\frac{1}{4}$ , 6–9 at  $\frac{3}{8}$ .

$\frac{3}{8}$  L

**(b) — instructor-judged.** Look for reasoning about *position*, not size. The median is whatever value sits in the middle place of the ordered list; adding one value at the right-hand end moves that middle place by one step only, and four beakers share the amount sitting there, so the value in the middle place does not change. A response saying only “because it is just one beaker” has not explained why the position matters.